Problem Statement

929 Pages of Documentation

533 Distinct Standards

27 Product Sections

Impossible to navigate without specialized expertise.

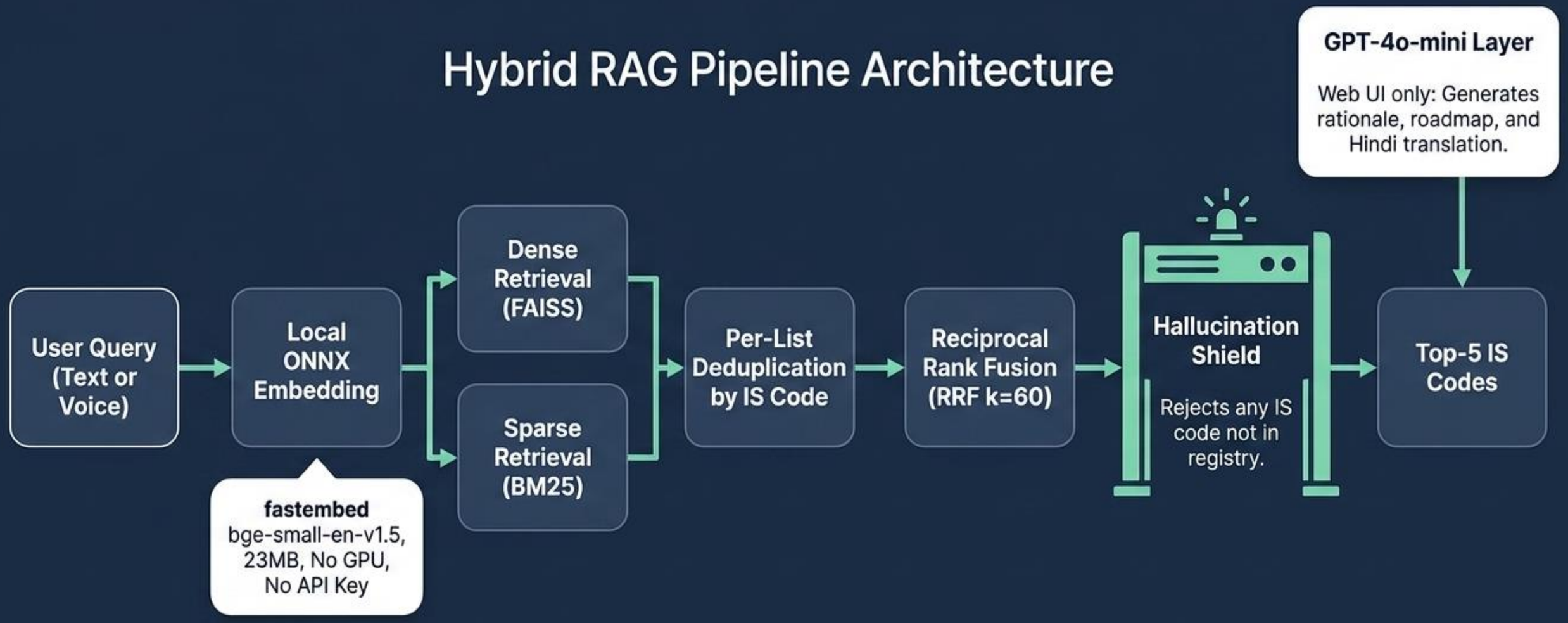
- Expensive consultants are the only viable path.
- Hindi-speaking manufacturers face strict language barriers.
- Weeks are wasted manually parsing irrelevant sections.

The Mission: Compress weeks of manual research into 0.02 seconds.

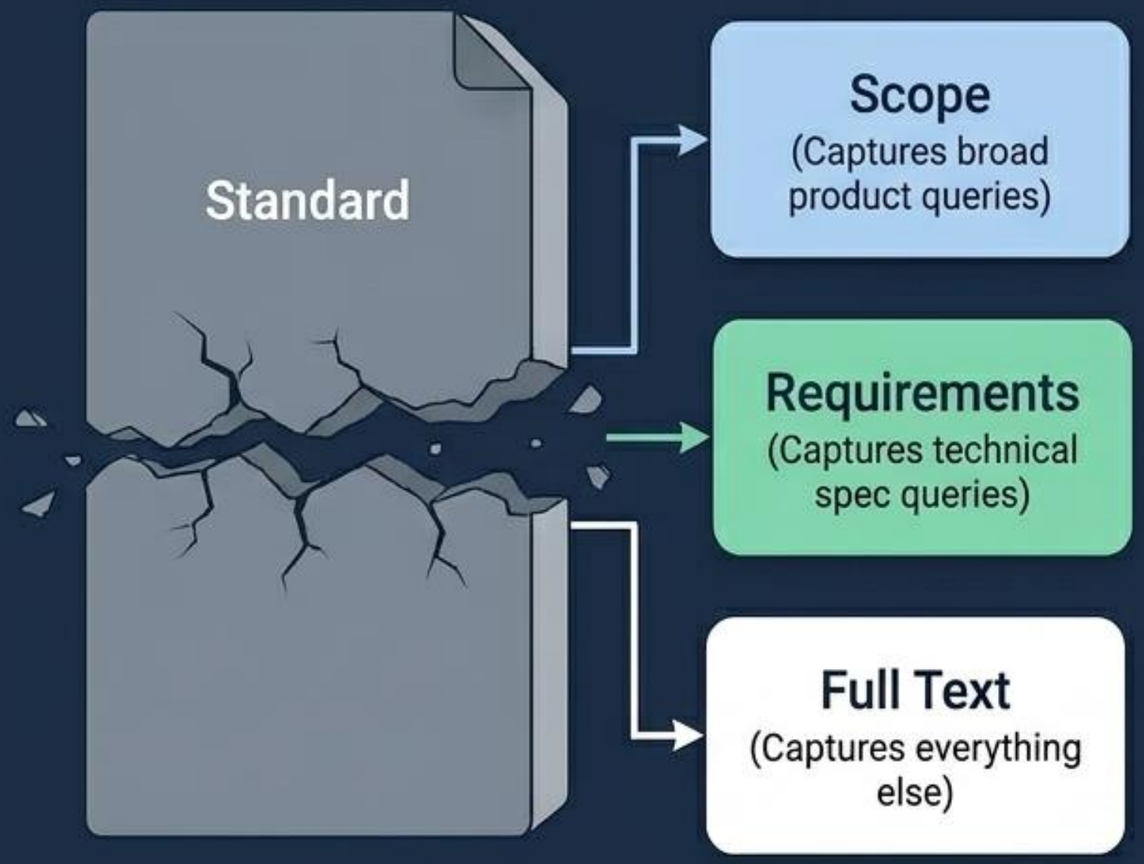
Solution Overview

929 pages. 533 standards. 0.02 seconds.

System Architecture



Chunking & Retrieval Strategy



1,602 total independent chunks indexed across 533 standards.

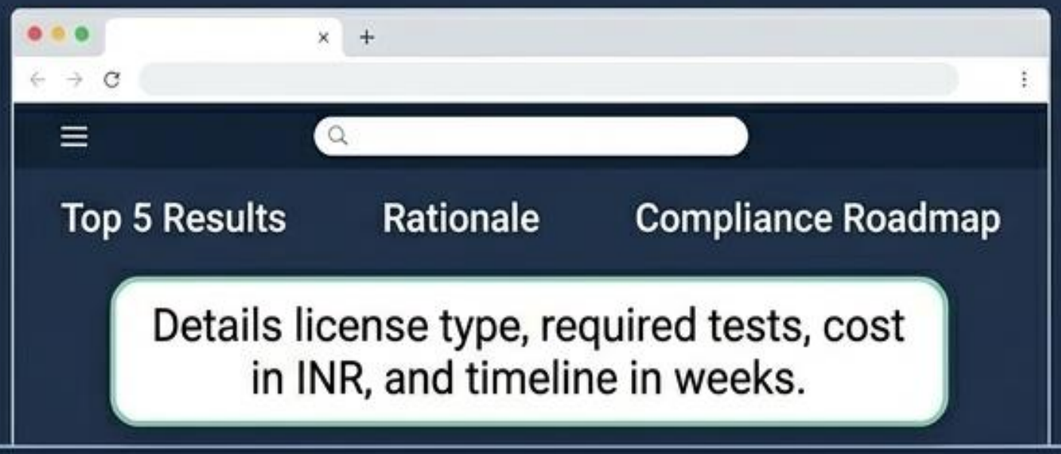
Architecture Upgrade Matrix

Standard Flat Indexing	Our Approach
• 1 flat document per standard	• 3 distinct semantic chunks
• Unfair scoring for fragmented docs	• Per-List Deduplication before RRF
• Hit Rate: ~70%	• **Hit Rate: 100%**

Key Innovation: Deduplication prevents standards with 3 chunks from getting an unfair 3x score advantage over 1-chunk standards.

Demo Highlights

Web UI Roadmap



Details license type, required tests, cost in INR, and timeline in weeks.

WhatsApp Voice Integration

Voice notes transcribed via Whisper. Queries executed via local retrieval only.

Knowledge Graph

Interactive exploration of 533 nodes and 600+ cross-reference edges.

Hindi Localization

Standard: IS 9876
Scope: Product testing
Requirements: Technical specs
Cost: 5000 INR

मानक: IS 9876
दायरा: उत्पाद परीक्षण
आवश्यकताएँ: तकनीकी विनिर्देश
लागत: 5000 INR

One-click full translation of all results for Hindi-speaking users.

Evaluation Results

100%

Hit Rate @3

Target: >80%. Status: Exceeded.

0.90

MRR @5

Target: >0.7. Status: Exceeded.

0.02s

Avg Latency

Target: <5s. Status: Exceeded (250x faster).

Hallucination Rate: 0%

Enforced by registry-based post-filter.

Fully Local Inference

Zero external API calls during scoring. Reproducible in 2 minutes on any machine.

Impact on MSEs

The Pain Point	The Co-pilot Reality
Weeks reading a 929-page PDF	✓ 0.02s query returns top-5 relevant standards
Cannot afford compliance consultants	✓ Free AI rationale explaining why each standard applies
No clarity on tests or licenses	✓ Compliance roadmap with exact INR cost estimates
Language barrier for Hindi speakers	✓ Full Hindi translation of all outputs
No PC or smartphone access required	✓ WhatsApp bot functions on basic devices via text or voice
Unclear standard interdependencies	✓ Visual knowledge graph mapping 533 standards and 600+ cross-references

Team & Acknowledgements

Team Name: Neura Rangers

Members: Saaheer Purav and Ojayit Telang

Event: Built for BIS x SS Hackathon, May 2026

Dataset Utilized: BIS SP 21:2005 (Provided by organizers)

Github: https://github.com/saaheerpurav/bis-rag

Live App: https://bis-rag.vercel.app

With thanks to the organizers, the Bureau of Indian Standards, and SS.